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Capstone Project Document

VISTA: A Vision-Language-Action System for Tabletop Manipulation with Robot Arms

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Capstone Project code	FA2AI63

- Ho Chi Minh city, 12/2026 -

Table of Contents

Acknowledgement 4

Definition and Acronyms 4

List of tables 4

List of figures 4

I. Project Introduction 5

1. Overview 5

2. Project Background 5

3. Project Objective 5

4. Problem Statement 5

5. Significance of the Project 5

6. Project Scope & Limitations 5

II. Project Management Plan 6

1. Team Work 6

2. Project Management Approach 6

III. Existing Systems/State of the Art 7

1. Overview of the Field 7

2. Historical Context 7

3. Key Studies and Theories 7

4. Technological Advancements 7

5. Comparison of Existing Systems 7

6. Gaps in the Literature/Technology 7

7. Justification for the Project 7

IV. Methodology 8

IV. Methodology (sample 1) 8

1. Research Questions and Objectives 8

2. Data Collection and Preprocessing 8

3. Feature Selection and Engineering 8

4. Model Training and Validation 8

5. Evaluation Metrics....8
6. Implementation Plan....8
7. Ethical Considerations....8
IV. Methodology (sample 2)....9
1. Project Objectives and RL Challenges....9
2. Environment Setup....9
3. Agent Design and States....9
4. Reward System....9
5. RL Algorithm....9
6. Training and Evaluation Strategy....9
7. Implementation Details....9
8. Ethical and Safety Considerations....9
V. System Design and Implementation....10
1. AI Model Integration....10
2. Data Flow and Processing....10
3. Deployment Strategy....10
4. Scalability and Maintenance....10
VI. Results and Discussion....11
1. Results and Analysis....11
2. Discussion....11
3. Recommendations....11
4. Personal Reflection (Optional)....11
VII. Conclusion....12
1. Summary of Findings....12
2. Contributions and Reflections on the Project....12
3. Limitations and Future Work....12
VIII. References....13
IX. Appendices....14

Acknowledgement

[Although not mandatory, a section thanking those who supported you during the project, including advisors, lecturers, friends, and family, can be included as an expression of gratitude.]

Definition and Acronyms

[A list of important terms and definitions used in the project. This ensures that all readers have a common understanding of the specialized terms used in the document.]

Acronym	Definition
AI	Artificial Intelligent
DL	Deep Learning
ML	Machine Learning
PM	Project Manager
VLA	Vision-language-action
VLM	Vision-language-model
ACT	Action-chunking-with-Transformers
RL	Reinforcement Learning
BC	Behaviour Cloning

List of tables

[A list of all the tables presented throughout the document.]

List of figures

[A list of all the figures presented throughout the document.]

I. Project Introduction

[Provide the final project introduction information following the template as part II in the Report #1]

1. Overview

1.1 Project Information

[Provide the brief information of the project: project name, project code, group name, list out the details of the project team and supervisor (and stakeholders), etc.]

Project name : VISTA: A Vision-Language-Action System for Tabletop Manipulation with Robot Arms

Project code : FA2AI63

Group name: GFA26AI04

Project teams:

+ Nguyễn Gia Phát - SE190236
+ Nguyễn Minh Đức - SE
+ Nguyễn Duy Sơn - SE
+ Nguyễn Lê Anh Quân - SE

Supervisor: Mr.Giang-Nam Luu, Ph.D

1.2 Project Overview

[A brief overview of the project, including the main parts that will be discussed in the document, to give readers a general view of the report's structure and content.]

This project investigates *"Whether replacing the original vision-language model (**VLM**) backbone of **SmolVLA** (Small Vision-Language-Action) with alternative lightweight VLMs can improve its performance in robotic manipulation while preserving its computational efficiency and accessibility?"*. The proposed approach focuses on evaluating alternative backbones with potentially stronger visual understanding capabilities and developing an efficient integration framework for incorporating them into the SmolVLA architecture. The central hypothesis is that an appropriately selected backbone can boost downstream manipulation performance without imposing disproportionate computational overhead.

To evaluate this hypothesis, experiments will be conducted using the low-cost SO-ARM101 robotic arm and the LeRobot framework. The modified SmolVLA variants (LMF2.5-VL-0.45B and FastVLM-0.5B as backbone replacement) will be compared against the original SmolVLA and Action Chunking with Transformers (**ACT**) as a baseline on two tabletop manipulation tasks: *Desk Stationery Decluttering* (pilot task) and *Conveyor Belt Object Sorting* (secondary task). The evaluation will examine task performance and computational efficiency to determine whether backbone replacement provides measurable benefits under practical hardware constraints.

This report presents the project from initial analysis to experimental evaluation. It begins with project planning and management, followed by an analysis of existing systems and relevant technologies. The proposed methodology and system design are then described, including backbone selection, integration,

data collection, and model training. Subsequently, the implementation and experimental evaluation procedures are presented, followed by an analysis and discussion of the results. Finally, the report summarizes the main findings, acknowledges the project's limitations, and identifies directions for future work.

The final deliverables of this project will be :

1. **Capstone Project Report :** A complete written report covering the whole project: the problem statement, the proposed solution (the backbone-swapping framework and the selected VLM variants), training and evaluation details, analysis of results, discussion, and recommendations. All reported results will be reproducible from the documented setup, and every claim will be checked against the evaluation data or a cited source.
2. **Demonstration Videos :** To show that the selected policy works on real hardware, we will record two videos of the SO-ARM101 performing the conveyor-belt sorting task: one with the best-performing SmolVLA variant and one with the original SmolVLA baseline for comparison. The videos will include the failure cases we observe, so the demonstration reflects actual performance rather than only the best runs.
3. **Control and Monitoring App (prototype) :** A simple application that lets users: run a chosen policy on the SO-ARM101, switch between policies (the baseline and the variants), and watch the live camera stream together with a 3D visualization of the arm's current pose.

2. Project Background

[Introduce the context and setting of the project. This may include why the project was chosen, which problem the project aims to address, and the importance of solving this issue. This helps the reader understand the cause and purpose of the project.]

Before 2023, robotic manipulation research was largely confined to large, expensive hardware, creating a significant barrier to entry for the broader community. The introduction of the ALOHA system and the ACT (Action Chunking with Transformers) paper revolutionized the field by demonstrating that fine-grained manipulation tasks could be achieved with high success rates on low-cost, imprecise hardware. This breakthrough catalyzed a wave of community-driven development, leading to the proliferation of affordable teleoperation systems, large-scale open datasets, and scalable learning methods. Key outcomes of this movement include the LeRobot library, an open-source framework for the entire robot learning stack, and the SO-ARM101, an even more accessible robotic arm. As hardware becomes more affordable, policies like ACT are being superseded by newer, more lightweight models such as SmolVLA, which are designed to perform manipulation tasks more effectively.

Despite being a compact and efficient policy, SmolVLA suffers from significant limitations. The first primary disadvantage, explicitly noted in Section 5.1 of its own paper, is its **choice of VLM backbone**. The model relies on an off-the-shelf VLM, SmolVLM, which is pretrained mainly on document reading and OCR tasks. Consequently, it is hypothesized that SmolVLM is not proficient at localization, describing, or identifying objects within images—a critical capability for robotic manipulation. This limitation arose from a design trade-off: the authors of SmolVLA aimed to create a small, efficient, and community-driven VLA that drastically reduces training and inference costs. Unlike previous VLAs, which are typically

massive (often with billions of parameters) and rely on academic or industrial datasets, SmolVLA's creators prioritized accessibility. As a result, their best and only option at the time was to use SmolVLM, the most compact and lightweight VLM available. These limitations are not necessarily inherent to SmolVLA's overall architecture; rather, they motivate an investigation into whether alternative lightweight VLM backbones can improve its visual understanding and downstream manipulation performance.

This design choice reflects a broader **trade-off between computational efficiency and model capability**. SmolVLA was developed to make VLA-based robotic control more accessible by reducing training and inference costs while leveraging compact pretrained components and an open-source development ecosystem. However, selecting a compact backbone may also constrain the representational capacity available for visually demanding manipulation tasks. In its discussion of limitations and future directions, the SmolVLA paper highlights the importance of exploring how these architectures can be scaled further without sacrificing speed or accessibility. This observation motivates a central question in the present project: **Can a carefully selected alternative VLM backbone improve manipulation performance while preserving, or further improving, SmolVLA's computational efficiency?**

As a result, this project focuses on two interconnected objectives: improving the visual-language capabilities of SmolVLA through backbone replacement and developing an efficient framework for integrating alternative VLMs into its architecture. Specifically, the project investigates whether replacing the original SmolVLM backbone with newer, lightweight VLMs can improve downstream robotic manipulation without undermining the efficiency and accessibility that characterize SmolVLA.

Since the release of the SmolVLA paper in June 2025, several new, reliable, and lightweight VLMs have been introduced that are better or even smaller than SmolVLM. These models are potential candidates for replacement. After careful consideration, we have selected LMF2.5-VL-0.45B and FastVLM-0.5B as the two suitable VLM models for testing and verifying the legitimacy of our solution, due to their lightweight characteristics and their inherent ability to process images effectively. By leveraging the low-cost SO-ARM101 hardware and the LeRobot library, we will evaluate our proposed solutions against the baselines (ACT and the original SmolVLA) on two tabletop manipulation tasks with different difficulty levels: Desk Stationery Decluttering and Conveyor Belt Object Sorting.

Importantly, this project is not limited to a straightforward plug-and-play replacement in which the original backbone is exchanged for another model without further adaptation. Instead, it aims to establish a systematic approach to backbone selection, integration, and evaluation, with the broader objective of improving the performance–efficiency trade-off of a lightweight VLA model. The investigation will examine the rationale for selecting the candidate backbones, compare their relevant architectural and computational characteristics, and develop an efficient integration framework for incorporating them into SmolVLA.

The subsequent sections describe the proposed framework, data collection procedures, model training configurations, experimental design, and evaluation metrics. The experimental results will then be analyzed to determine whether backbone replacement produces meaningful improvements in manipulation performance and computational efficiency, and to identify the conditions under which these improvements can be achieved.

3. Project Objective

[The specific objectives of the project. This includes what the project hopes to achieve at the highest level. Objectives should be clear, measurable, and achievable. This guides the development of the project and the evaluation of its success.]

The primary objective of this project is to investigate whether replacing SmolVLA's original vision-language model (VLM) backbone with alternative lightweight VLMs can improve robotic manipulation performance while maintaining or enhancing computational efficiency. This investigation is motivated by two related challenges: the suitability of the original backbone's visual representations for manipulation tasks and the potential trade-off between model capability, computational cost, and scalability.

To address this, the project establishes the following specific objectives:

1. **Design and implement alternative backbone architectures.** Adapt and integrate LFM2.5-VL-0.45B and FastVLM-0.5B into the SmolVLA architecture as alternative VLM backbones. Develop the necessary integration and compatibility mechanisms, and produce *two trainable model variants* suitable for subsequent experimentation.
2. **Train the baseline and modified models.** Train and evaluate four model configurations: the original ACT baseline, the original SmolVLA baseline, SmolVLA with the LFM2.5-VL-0.45B backbone, and SmolVLA with the FastVLM-0.5B backbone. Record *training logs and relevant metrics*, including training loss, training duration, step execution time, and other available indicators of computational cost, to support reproducibility and comparative analysis.
3. **Evaluate manipulation performance and computational efficiency.** Conduct comparative experiments on the *Desk Stationery Decluttering* and *Conveyor Belt Object Sorting* tasks. Measure task success rates and relevant efficiency metrics, including inference latency, memory consumption, and model parameter count where applicable. Analyze the results to determine whether backbone replacement provides measurable improvements over the original SmolVLA and *identify the best-performing configuration* under the defined evaluation criteria.
4. **Demonstrate real-world applicability.** Deploy the selected model on the SO-ARM101 robotic platform and investigate its suitability for a real-world goods-arrangement task involving a mobile conveyor line at a partner factory producing packaged coffee. Produce a *video demonstration documenting* the system's operation in the target environment and assess its practical limitations under real-world operating conditions.
5. **Develop a reusable backbone-integration strategy.** Document the architectural modifications, compatibility requirements, implementation procedures, and evaluation considerations involved in replacing SmolVLA's original VLM backbone. Based on the experimental findings, *formulate a systematic integration strategy* that can inform future investigations of alternative backbones and potentially support adaptation to other VLA architectures.
6. **Assess the potential for improving new lightweight VLA models (Highest-level).** Determine whether the proposed backbone modifications achieve a favorable balance between manipulation performance and computational efficiency. Compare the resulting models with the

selected baselines and discuss the extent to which the *findings support further development of lightweight, accessible VLA systems.*

4. Problem Statement

[A specific statement of the problem that the project intends to solve. This section should detail the issue, including its causes, impacts, and specific challenges associated with solving the problem]

SmolVLA is a compact vision-language-action (VLA) policy of about 0.45B parameters. It is designed to be trained on a single GPU and deployed on consumer GPUs or even CPUs, while staying competitive with much larger VLAs on manipulation tasks. Despite this, the authors identify limitations that this project targets.

Limitation 1: an off-the-shelf backbone that is not suited to robotics. SmolVLA uses SmolVLM-2 as its backbone. SmolVLM-2 is a general-purpose VLM pretrained mainly on document understanding and OCR. Whether such a backbone is a good starting point for spatial relationships, affordances, and physical interaction is an open question the authors raise themselves. SmolVLA also keeps its visual input minimal: it disables image tiling and uses only 64 visual tokens per frame, which may sacrifice fine detail on small objects. Together, these choices make perception and object localization in cluttered scenes a plausible weak point.

Limitation 2: little room to improve without adding weight. The authors list further model-size scaling as future work. SmolVLA already uses a very small VLM and only a subset of its early layers. Gains from the backbone therefore cannot come from simply making the model larger without giving up the efficiency that defines it.

Why do these limitations exist? When SmolVLA was released (June 2025), SmolVLM was the best accuracy-per-memory option among small VLMs. Its smaller variants need roughly 0.8–1.2 GB of RAM at batch size 1, compared with about 27.7 GB for *MolmoE-A1B-7B*, and it matched or beat comparable models such as *InternVL2-2B* on several benchmarks. Under a tight efficiency budget, there was little reason to choose another backbone. Because the architecture was then tuned around this backbone, it left little room for change.

What has changed. Newer compact VLMs now offer a possible alternative: FastVLM (CVPR 2025, 0.5B variant), LFM2-VL (450M and 1.6B), and Qwen3.5-0.8B (released in early 2026). Their training data and architectures differ from SmolVLM's, so they may provide better visual grounding at a similar scale.

Problem statement. *Can carefully selected alternative VLM backbones improve the manipulation performance of SmolVLA while preserving its computational efficiency and practical deployability?*

Challenges.

1. *Backbone selection.* Candidates must stay lightweight, and their pretraining should emphasize natural-image understanding and grounding rather than documents.

2. *Layer selection.* SmolVLA uses only part of its VLM's layers, so the best layers to keep must be re-determined for each new backbone. Some candidates use non-standard layer types (for example, LFM2-VL's hybrid convolution-attention backbone), which complicates this.
3. *Integration.* There is no ready-made, tested way to swap the backbone, so we must build the adapter and training pipeline ourselves, covering token and feature dimensions, the action-expert connection, and the training setup.
4. *Uncertain outcome.* A new backbone may not remove either limitation. Conclusions will rest on training and evaluating each variant against the original SmolVLA, in simulation and on real tasks, measuring both success rate and efficiency (parameters, memory, latency).

5. Significance of the Project

[The importance of the project. This is where you describe the benefits that the project brings, both theoretically and practically, and its impact on the scientific community, industry, or society at large.]

SmolVLA was created to make vision-language-action (VLA) research more accessible. Most VLAs have billions of parameters and are trained on academic or industrial datasets. SmolVLA is a compact policy trained on community-collected data. It can be trained on a single GPU and deployed on consumer hardware, while remaining competitive with much larger models on manipulation tasks. The limitations described in Section 4 (a backbone pretrained mainly on documents and OCR, and little room to improve without adding weight) restrict how far this approach can go. This project addresses those limitations, and its significance lies in three areas.

Scientific significance. The project provides an empirical answer to an open question raised by SmolVLA's authors: whether a VLM backbone better suited to visual grounding than the original one improves robotic manipulation at a comparable scale. Whatever the outcome, the results are informative. Gains would show that backbone choice matters in lightweight VLAs.

Practical and industrial significance. The project is implemented and evaluated on the low-cost SO-ARM101 robot arm through a conveyor-belt object-sorting task. If the improved variants perform reliably, the results will suggest that affordable hardware paired with a compact policy can handle simple, repetitive sorting workflows. That would give small and medium-sized factories a cheaper option than expensive industrial robots.

Community and methodological significance. The most reusable result is the framework itself: a documented strategy for choosing a replacement backbone, selecting which of its layers to use, and adapting it into SmolVLA for training and evaluation. Today, swapping a backbone in a VLA requires custom engineering each time. A tested procedure lowers that barrier. Researchers with stronger hardware can apply the same process to larger VLM backbones, and those with limited resources can test small ones. It also helps keep progress on lightweight VLAs open to people without large GPU clusters.

6. Project Scope & Limitations

[The scope of the project as well as its limitations. This includes what the project will and will not cover. This section helps set expectations and prevent misunderstandings.]

What the project will cover : (1) Backbone replacement. We will replace SmolVLA's SmolVLM-2 backbone with a small number of lightweight VLMs. The candidates are drawn from the models in Section 4 (Qwen3.5-0.8B, FastVLM, and LFM2-VL or its newer LFM2.5-VL version). The final list will depend on how well each can be integrated and how much compute is available. **(2) Swapping framework.** We will build and document a procedure for swapping the backbone. It covers choosing the layers to keep, matching feature dimensions to the action expert, and setting up training so that each variant can be trained and tested. **(3) Data collecting.** As for the calibration on each hard-ware is different due to human stochastic in action and we do not use any stimulations, we will follow the design in experiment protocol of SmolVLA paper to collect 300 demonstrations for each task "*Desk Stationery Decluttering"* (pilot task) and "*Conveyor Belt Object Sorting"* (secondary task). **(4) Training.** Each variant and the original SmolVLA baseline will be trained under the same data and settings, so the comparison is fair. **(5) Evaluation.** The variants and baseline will be compared on the desk decluttering and conveyor-belt object-sorting task using the SO-ARM101 arm. We will measure task success rate. We will also measure efficiency: parameter count, memory use, and inference latency. **(6) Analysis.** We will analyze which backbone properties (pretraining data, vision encoder, layer choice) appear to relate to performance differences, and report limitations of the framework.

What the project will not cover : (1) Designing a new VLA architecture or modifying the action expert, other than the changes needed to connect a new backbone. **(2)** Large-scale pretraining on community datasets, or training backbones from scratch. **(3)** Cross-embodiment evaluation, long-horizon tasks, or tasks other than conveyor-belt sorting. **(4)** Larger VLM backbones (>1B parameter), or any model where efficiency requirements no longer apply. **(5)** Deployment engineering for production use, such as safety certification, long-term reliability testing, or integration with factory systems. **(6)** Fully eliminating the two limitations in Section 4. We aim to measure how much backbone replacement helps, not to prove it solves them.

Limitations of this project :

- **Small candidate set.** Only a few backbones can be tested, so a better-suited model may exist that we do not try.
- **Limited data and trials.** Training and evaluation data will be small compared with the original SmolVLA pretraining data, and real-world trial counts will be limited by time.
- **Single task and robot.** Results come from one task on the low-cost SO-ARM101 arm, whose payload and precision are limited. They indicate feasibility for light sorting tasks only.
- **Compute limits.** Hyperparameter tuning and training length are restricted by available GPU resources, so reported performance may not reflect each backbone's best possible result.
- **Fast-changing field.** Newer lightweight VLMs may appear during the project, and the candidate list will be fixed at a defined point.

II. Project Management Plan

[Provide the final project plan information following the template as part II in the Report #2]

1. Team Work

1.1 Team Structure and Roles

[Describe the structure of the team and the roles of each member. This should include information about specific tasks, responsibilities, and communication methods between members.]

Member	Role	Responsibilities
Nguyễn Gia Phát	Project Manager	Divides tasks, builds and updates the plan, tracks milestones, and leads research on the proposed method. Main contact for the supervisor and the factory partner. Compiles the final report.
Nguyễn Minh Đức	AI Engineer	Trains the policies, implements the backbone-swapping method in code, and develops the control and monitoring app.
Nguyễn Lê Anh Quân	AI Engineer	Reviews and tests the code, tunes policy hyperparameters, maintains the result logs, analyses metrics for insights, and supports app testing.
Nguyễn Duy Sơn	Data Engineer	Collects, cleans, and annotates data for both tasks. Prepares the test setup (robot, camera, task layout), runs the real-robot trials with Quân, records the demonstration videos, and supports research and verification of proposals.

Every member writes the report sections that cover their own work, and Phát compiles and checks the final document. The team uses Zalo, Gmail, and Discord. In-person meetings are held twice a week, on Wednesday and Sunday. Each member reports completed work within the same day so progress can be checked daily

1.2 Communication Plan

[Describe how and how often project information will be communicated among team members and advisor (and other stakeholders).]

Who	How often	Purpose
Project Manager and team	Daily (end of day)	Check that each member's tasks are done and spot problems early.
Whole team	Twice a week (Wednesday, Sunday)	Held direct, in-person meeting. Review deliverables and milestones, and adjust the plan in time.
AI Engineers	Daily	Record work in the worklog. Report to the Project Manager when a training run finishes, or immediately when a problem requires a change to the plan.
Data Engineer	Daily	Report collection and cleaning progress in the daily report. Notify the AI Engineers when a dataset version is ready, and the Project Manager if collection is blocked or data quality affects the schedule.
Factory partner	Once a week	The Project Manager checks for changes or requests and confirms the testing schedule. Before on-site testing, the team confirms site access, safety rules, and permission to record video. New requests follow the change-control rule in Section 6.
Supervisor	Once a week	Ask for advice on ideas, plans, and methods, report progress, and take feedback. The supervisor also decides, together with the team, whether a new factory request fits the scope.

2. Project Management Approach

2.1 WBS

[Provide detailed descriptions for each task or duty within the WBS, including objectives, required resources, and assigned responsibilities.]

WBS ID	Work package / task	Objective & description	Deliverable / evidence to attach	Required resources	Owner	Support	Start	Days	End	Pred.	Report Obj.
1	**Project management & governance**	**Plan, track and control the project; keep evidence for every task.**					**15-Sep**	**77**	**30-Nov**		
1.1	Kickoff, roles, tools & repository setup	Agree roles and communication rules; create the Git repo, shared drive, worklog and meeting-minute templates.	Role table; repo and worklog templates	GitHub, Zalo, Gmail, Discord	Phát	All	15-Sep	4	18-Sep	-	PM
1.2	Project plan: WBS, schedule & risk register	Break the work into work packages with owners, dates and deliverables; identify the top risks.	This WBS; risk register; Report Section II	Excel, report template	Phát	All	15-Sep	8	22-Sep	1.1	PM
1.3	Progress tracking & meetings	Daily worklog; team meetings (Wed, Sun); weekly supervisor and factory updates. Attach evidence (logs, photos, run IDs) to every finished task.	Worklog; meeting minutes; evidence log	Worklog, Discord	Phát	All	15-Sep	77	30-Nov	1.1	PM
1.4	Risk & change-control log	Update the risk register weekly; log every scope or schedule change with its impact and the supervisor's decision.	Risk register; change log	Excel	Phát	Quân	22-Sep	70	30-Nov	1.2	PM

1.5	Project closure & lessons learned	Compare plan vs actual, record lessons learned, archive code, data and results.	Closure summary; archived repo and data	Git, shared drive	Phát	All	24-Nov	7	30-Nov	10.12	PM
2	**Research & requirements**	**Define the task and build the literature base for the study.**					**15-Sep**	**24**	**08-Oct**		
2.1	Task definition & factory requirements	Specify the sorting task: which objects, how many, layout, belt speed, lighting, success criteria. Confirm with the partner factory.	Task specification sheet; partner confirmation	Factory contact, SO-ARM101 reach data	Phát	Sơn	15-Sep	10	24-Sep	-	O4
2.2	Literature search & screening	Run a documented search (sources, queries, inclusion/exclusion) and build the reference library.	Search log; reference library; screened paper list	arXiv, Google Scholar, Zotero	Phát	Sơn	15-Sep	12	26-Sep	-	O6
2.3	Architecture study: ACT, SmolVLA, LeRobot, SO-ARM101	Analyse prior architectures: inputs and outputs, data flow, method, reported results, strengths and limits.	Architecture notes with diagrams, formulas, sources	Papers, LeRobot code	Đức	Quân	17-Sep	10	26-Sep	-	O1
2.4	Candidate backbone survey & selection	Compare lightweight VLMs on size, pretraining data, layer structure, licence and speed. Shortlist LFM2.5-VL-0.45B and FastVLM-0.5B.	Selection table with criteria and decision	Hugging Face model cards, papers	Đức	Quân	24-Sep	10	03-Oct	2.3	O1
2.5	Concept matrix & gap analysis (Section III draft)	Compare existing systems on common criteria and derive the research gap from the matrix.	Concept matrix; Section III draft	Reference library	Phát	Đức	29-Sep	10	08-Oct	2.2, 2.4	O6
3	**Methodology & experimental design**	**Fix research questions, metrics, baselines and the evaluation protocol before the final runs.**					**24-Sep**	**19**	**12-Oct**		

3.1	Research questions, hypotheses & success criteria	State the research question, hypothesis and numeric pass criteria for each objective.	RQ and hypothesis list; success-criteria table	-	Phát	Quân	24-Sep	7	30-Sep	2.1	O6
3.2	Metrics selection & justification	Choose success rate, progress score, latency, memory and parameter count. Give formulas and explain why other metrics are unsuitable.	Metrics specification with formulas	Papers	Quân	Phát	29-Sep	7	05-Oct	3.1	O3
3.3	Baseline selection & reproduction	Set up ACT and original SmolVLA in LeRobot (at least 2 baselines) and compare with numbers published in other studies.	Baseline configs; comparison table vs published results	LeRobot, Kaggle GPU	Đức	Quân	29-Sep	12	10-Oct	2.3	O2
3.4	Method rationale: formulas, citations, noise space	Document the formulas of the chosen method, including the flow-matching noise space: why it is used and how it applies in the real setting.	Method rationale note with formulas and sources	SmolVLA and flow-matching papers	Đức	Phát	01-Oct	8	08-Oct	2.3	O1
3.5	Evaluation protocol & statistics plan	Define trial counts (with sample-size reasoning), randomized start positions, success definitions, confidence intervals and tests.	Evaluation protocol document	Python (SciPy, statsmodels)	Quân	Phát	03-Oct	9	11-Oct	3.1, 3.2	O3
3.6	◆ Protocol freeze	Protocol and metrics frozen before the final trials.	Dated, versioned protocol	-	Phát	All	12-Oct	1	12-Oct	3.5	O3
4	**Data collection & preprocessing**	**Build a documented, diverse dataset for both tasks.**					**22-Sep**	**39**	**30-Oct**		

4.1	Hardware setup & calibration	Set up leader/follower SO-ARM101, cameras and task layouts; calibrate and record the setup.	Calibration files; setup photos; layout diagram	SO-ARM101, cameras, conveyor mock-up	Sơn	Quân	22-Sep	10	01-Oct	2.1	O2
4.2	Dataset specification & collection protocol	Define data source (self-collected vs public), number of samples, features, variables X/Y/Z, storage format and diversity plan.	Dataset datasheet; collection protocol	LeRobot dataset format	Sơn	Quân	29-Sep	7	05-Oct	4.1	O2
4.3	Pilot collection & quality check	Collect pilot episodes; check synchronization, image quality and episode lengths; fix the protocol.	Pilot report with sample videos	SO-ARM101, cameras	Sơn	Đức	05-Oct	5	09-Oct	4.2	O2
4.4	Collect demonstrations – Task A (Desk Stationery Decluttering)	Teleoperate and record the target of 300 demonstrations with varied object positions.	Task A dataset; collection log	SO-ARM101, cameras	Sơn	Quân	10-Oct	8	17-Oct	4.3	O2
4.5	Collect demonstrations – Task B (Conveyor Belt Object Sorting)	Teleoperate and record the target of 300 demonstrations with varied object positions and belt conditions.	Task B dataset; collection log	SO-ARM101, cameras, conveyor mock-up	Sơn	Quân	18-Oct	8	25-Oct	4.4	O2
4.6	Data cleaning, QC, episode-level split & statistics	Remove failed or corrupt episodes, split by episode, compute dataset statistics (size, depth, breadth, diversity).	Cleaning report; split files; statistics tables	Python	Sơn	Quân	14-Oct	16	29-Oct	4.4	O2
4.7	◆ Dataset v1.0 frozen	Dataset version locked before any final training run.	Versioned dataset with hash	-	Sơn	Quân	30-Oct	1	30-Oct	4.6	O2
5	**Model integration (backbone replacement)**	**Integrate the alternative VLM backbones into SmolVLA.**					**01-Oct**	**50**	**19-Nov**		

5.1	Integration design	Design the adapter: token and feature dimensions, which layers to keep for each backbone, connection to the action expert.	Integration design doc with diagram	LeRobot code	Đức	Quân	01-Oct	10	10-Oct	2.4, 3.4	O1
5.2	Implement LFM2.5-VL-0.45B variant	Implement the backbone and adapter in LeRobot; handle its hybrid convolution-attention layers.	Working code branch; config file	LeRobot, Hugging Face	Đức	Quân	08-Oct	12	19-Oct	5.1	O1
5.3	Implement FastVLM-0.5B variant	Implement the backbone and adapter in LeRobot.	Working code branch; config file	LeRobot, Hugging Face	Đức	Quân	11-Oct	12	22-Oct	5.1	O1
5.4	Sanity tests, parameter count & complexity	Forward-pass and small-subset overfit tests; count parameters (millions), estimate complexity, assess ways to reduce parameters.	Test log; parameter and complexity table	PyTorch profiler	Quân	Đức	20-Oct	8	27-Oct	5.2, 5.3	O1
5.5	Reusable integration strategy documentation	Write the swap procedure, compatibility checklist and lessons for future backbones.	Integration guide; material for Sections V and VII	-	Đức	Phát	10-Nov	10	19-Nov	5.4	O5
6	**Training**	**Train baselines and variants under identical data and settings.**					**20-Oct**	**32**	**20-Nov**		
6.1	Training configuration & hyperparameter justification	Set steps, batch size, learning rate and seeds; justify each choice from papers or calculations; plan Kaggle GPU sessions.	Config table with sources; Kaggle session plan	Kaggle GPU	Quân	Đức	20-Oct	8	27-Oct	3.3, 5.1	O2
6.2	Train baselines: ACT & original SmolVLA	Train on the frozen dataset with the common settings; log steps, runtime and GPU.	Checkpoints; training logs	Kaggle GPU	Quân	Đức	31-Oct	9	08-Nov	6.1, 4.7	O2

6.3	Train SmolVLA + LFM2.5-VL-0.45B	Train with the same data and settings as the baselines.	Checkpoints; training logs	Kaggle GPU	Đức	Quân	01-Nov	12	12-Nov	6.1, 4.7, 5.4	O2
6.4	Train SmolVLA + FastVLM-0.5B	Train with the same data and settings as the baselines.	Checkpoints; training logs	Kaggle GPU	Đức	Quân	03-Nov	12	14-Nov	6.1, 4.7, 5.4	O2
6.5	Fine-tuning improvement round	Try tuning directions on the best variant (for example layers kept, learning-rate schedule, image tokens); record before/after metrics for each direction.	Before/after comparison table	Kaggle GPU	Quân	Đức	13-Nov	8	20-Nov	6.3, 6.4	O2
6.6	Training log & reproducibility pack	Collect loss curves, steps, runtime, GPU configuration, seeds and commit hashes for every run.	Reproducibility pack	local logs, Git	Quân	Đức	05-Nov	16	20-Nov	6.2	O2
6.7	◆ Training complete	All final checkpoints available.	Checkpoint list	-	Quân	Đức	20-Nov	1	20-Nov	6.5	O2
7	**Evaluation**	**Measure manipulation performance and efficiency with the frozen protocol.**					**09-Nov**	**19**	**27-Nov**		
7.1	Offline validation on held-out episodes	Compute action-prediction error on held-out episodes for all four models as an early quantitative signal.	Offline results table	Python	Quân	Đức	09-Nov	7	15-Nov	6.2	O3
7.2	Efficiency benchmarking	Measure latency (median and p95, warm-up excluded), memory use and parameter count on the target hardware.	Efficiency table	Target PC/GPU, profiler	Quân	Đức	13-Nov	5	17-Nov	6.3, 6.4	O3

7.3	Real-robot trials: baselines	Run the planned trials for ACT and original SmolVLA; log every trial and record video.	Trial log; videos	SO-ARM101, cameras	Sơn	Quân	09-Nov	7	15-Nov	6.2, 3.6	O3
7.4	Real-robot trials: backbone variants	Run the planned trials for both variants under the same protocol; log every trial and record video.	Trial log; videos	SO-ARM101, cameras	Sơn	Quân	14-Nov	8	21-Nov	6.3, 6.4, 3.6	O3
7.5	Re-evaluate the tuned best model	Repeat the trials for the model from the fine-tuning round.	Trial log; videos; before/after results	SO-ARM101, cameras	Sơn	Quân	21-Nov	5	25-Nov	6.5	O3
7.6	Statistical & failure-case analysis	Compute confidence intervals and tests; classify failures (perception, grasp, timing, hardware); select qualitative cases.	Statistics tables; failure taxonomy; qualitative examples	Python	Quân	Phát	22-Nov	6	27-Nov	7.3, 7.4	O3, O6
8	**Control & monitoring app (prototype)**	**Let users run a policy, switch between policies and watch the camera plus 3D arm pose.**					**12-Oct**	**32**	**12-Nov**		
8.1	App requirements & architecture	List functions, interfaces and the data flow between app, policy and robot.	Requirements list; architecture diagram	-	Đức	Phát	12-Oct	6	17-Oct	2.3	O4
8.2	Policy runner & policy switching	Run a chosen policy on the SO-ARM101 and switch between baseline and variants.	Working module	Python, LeRobot	Đức	Quân	23-Oct	10	01-Nov	8.1, 5.4	O4
8.3	Live camera stream & 3D arm-pose visualization	Show the live camera feed together with the arm's current pose in 3D.	Working module	Python, 3D viewer library	Quân	Đức	28-Oct	10	06-Nov	8.1	O4

8.4	App integration test on SO-ARM101	Test the whole app on the real robot; fix defects.	Test report; app prototype	SO-ARM101, cameras	Quân	Đức	07-Nov	6	12-Nov	8.2, 8.3	O4
9	**Demo & real-world deployment**	**Show the selected policy on real hardware, including failure cases.**					**12-Nov**	**17**	**28-Nov**		
9.1	Demo setup & safety check	Check emergency stop, joint limits and workspace; confirm site access and recording permission.	Safety checklist; permission confirmation	SO-ARM101, factory contact	Sơn	Phát	12-Nov	4	15-Nov	2.1	O4
9.2	Demo dry-run for Review 3 & fixes	Run the full demo with the best model so far; fix the problems found.	Dry-run video; defect list	SO-ARM101, app	Quân	Sơn	16-Nov	4	19-Nov	8.4, 9.1	O4
9.3	On-site factory trial & demo videos	Record the best variant and the original SmolVLA on the conveyor-belt task, including failure cases.	Two demo videos; on-site test notes	SO-ARM101, factory conveyor, camera	Sơn	Phát, Quân	25-Nov	4	28-Nov	7.5, 9.1	O4
10	**Reporting, reviews & submission**	**Write and verify the report; prepare the Review 2 and Review 3 evidence.**					**06-Oct**	**56**	**30-Nov**		
10.1	Report: Sections I–II (final)	Final introduction and project management plan; remove inconsistencies in scope, tasks and models.	Sections I–II	Report template	Phát	All	06-Oct	14	19-Oct	1.2, 3.1	PM
10.2	Report: Section III – existing systems	Write literature review from the concept matrix and search log.	Section III	Reference library	Phát	Đức	08-Oct	10	17-Oct	2.5	O6
10.3	Report: Section IV – methodology	Write research design, data, training setup, protocol, statistics and threats to validity.	Section IV	Report template	Quân	Phát	12-Oct	14	25-Oct	3.6, 4.2	O2, O3

10.4	Review 2 evidence pack & slides	Collect roadmap, dataset specification, architecture analysis, baselines, metrics and current numbers.	Review 2 slides; evidence folder	Slides, evidence log	Phát	All	20-Oct	7	26-Oct	1.3	PM
10.5	◆ Review 2 (27/10/2026)	Review meeting with the reviewer.	Review 2 form with comments	-	Phát	All	27-Oct	1	27-Oct	10.4	PM
10.6	Review 2 feedback → action plan & WBS update	Turn each reviewer comment into tasks, owners and dates; update this WBS and the risk register.	Action list; updated WBS	Excel	Phát	All	28-Oct	3	30-Oct	10.5	PM
10.7	Report: Section V – system design	Write architecture, data flow, interfaces, deployment and latency budget.	Section V	Report template	Đức	Quân	03-Nov	10	12-Nov	5.4, 8.1	O1, O4
10.8	Review 3 evidence pack & slides	Collect results tables, failure cases, tuning before/after and demo video.	Review 3 slides; evidence folder	Slides, evidence log	Phát	All	14-Nov	5	18-Nov	7.4	PM
10.9	◆ Review 3 (date to be confirmed)	Review meeting with the reviewer. Date is a placeholder until confirmed.	Review 3 form with comments	-	Phát	All	20-Nov	1	20-Nov	10.8	PM
10.1	Report: Section VI – results & discussion	Present results with n and confidence intervals; give each objective a verdict; discuss limits and threats.	Section VI	Report template	Phát	Quân	19-Nov	9	27-Nov	7.4, 7.6	O3, O6
10.11	Report: Conclusion, references, appendices, front matter	Write Section VII, finalize references, appendices, abstract, acronym list, lists of tables and figures.	Sections VII–IX; front matter	Zotero, report template	Phát	Sơn	23-Nov	6	28-Nov	10.10	PM

10.12	Claim-ledger verification & quality gates (G1–G3)	Check every claim against its source or run log; a different person verifies than the one who wrote it.	Verified claim ledger; gate checklist	Spreadsheet ledger	All	Phát	22-Nov	7	28-Nov	10.7, 10.10	PM
10.13	Final proofreading & report freeze	Language, formatting and number-consistency check.	Final report PDF/DOCX	Report template	Phát	All	28-Nov	2	29-Nov	10.11, 10.12	PM
10.14	*Final submission*	Submit the report, videos and app.	Submitted package	-	Phát	All	30-Nov	1	30-Nov	10.13	PM

2.2 Risk Management

[Describe potential risks during the project execution and preventive measures or response plans for each risk.]

Risk management is conducted through the identification, assessment, mitigation, and monitoring of potential events that could affect the project's schedule, resources, or deliverables. The risks are evaluated according to their likelihood of occurrence and potential impact on project completion. Preventive measures are established to reduce the probability of occurrence, while contingency plans are prepared to minimize disruption if a risk materializes. Three major risks have been identified for this project:

Risk 1: Training delays and loss of experimental results

Training the selected Vision-Language-Action (VLA) models and their variants may require substantial computational resources and execution time. Training may be delayed by limited GPU availability, software or hardware errors, interrupted sessions, or unexpected termination before model checkpoints and experimental results are saved. Such incidents could delay subsequent activities, including evaluation, comparison, and report preparation.

To mitigate this risk, the team will use GPU-enabled Kaggle environments, subject to available computing resources and usage quotas. Both AI engineers will prepare their training environments and execute independent experiments in parallel where resource availability permits. Training configurations, logs, evaluation results, and model checkpoints will be saved regularly and copied to persistent storage.Training will be prioritized according to the project's experimental objectives, with sufficient time reserved for evaluation and analysis. If the available compute budget becomes insufficient, the team will prioritize the essential baseline and primary experimental conditions before conducting supplementary experiments.

Risk 2: SO-ARM101 hardware failure

The project depends on the SO-ARM101 robotic platform for data collection and physical evaluation. Hardware failure caused by component wear, mechanical damage, electrical faults, or accidental mishandling could interrupt these activities. Since replacement components or a replacement device may require international shipping, equipment downtime could substantially delay the project.

To reduce this risk, the team will inspect and test the robotic platform before data collection and evaluation, follow appropriate operating procedures, and maintain backups of the collected datasets and associated configurations. A second SO-ARM101 arm will be prepared as a replacement, subject to successful assembly and compatibility testing. If the primary arm fails, the team will switch to the backup platform and verify its operation before resuming experiments. If both platforms become unavailable, the team will prioritize analysis of previously collected data, model training, software debugging, and other tasks that do not require physical access to the robot. Physical evaluation will resume when the equipment becomes available, and any resulting limitations will be documented in the final report.

Risk 3: Team member unavailability

A team member may become temporarily or permanently unavailable because of illness or other unforeseen circumstances. Since the project involves interdependent activities, including data collection, model training, evaluation, and report preparation, the absence of a member responsible for a critical task could delay dependent activities and create an uneven workload among the remaining members.

To mitigate this risk, the team will document task procedures, experimental configurations, data locations, and implementation details in a shared project repository. Each critical task will have a designated backup member who is familiar with its essential procedures. Team members will maintain sufficient cross-functional knowledge to take over urgent tasks when necessary. If a member becomes unavailable, the team leader will reassess task priorities, redistribute responsibilities according to each member's capacity and expertise, and adjust the schedule where necessary. Critical deliverables, particularly data collection, baseline experiments, evaluation, and report completion, will receive priority over nonessential extensions.

These risks will be monitored throughout the project. If a risk materializes or its likelihood increases, the team will assess its effect on the schedule and deliverables, implement the corresponding contingency plan, and document any resulting changes to the project plan. This approach is intended to reduce avoidable delays and preserve the feasibility of completing the project's essential objectives within the available time and resources.

2.3 Quality Management

[Details on the measures and processes that will be applied to ensure the quality of the project, including testing and validation of the product or project outcomes.]

2.4 Budget and Funding (Optional)

[If applicable, describe the project budget and funding sources. This should include details on estimated costs and financial management throughout the project.]

The project had no external funding for hardware. The team members self-funded the physical setup. Recorded spending from July to October 2026 was **12,504,000 VND** (about 3,126,000 VND per member), listed in Table X.

Table X. Hardware expenditure (thousand VND)

Category	Description	Cost
Servos	2 sets of 12 servos for SO-ARM101 (leader and follower arms), including visa/transfer fees. The second set supports the backup arm in Risk 2.	9,385
Control electronics	4 bus driver boards; 2 × 6V adapters	1,114
Cameras	3 camera modules and 1 additional camera	881
3D-printing material	3 rolls of PETG for the arm's shell	352
Shipping fee	China to Vietnam	105
Accessories	USB hubs, Type-C cables, tape, storage box, camera stand	667
Total		**12,504**

The partner lab sponsors the project by allowing the team to use its 3D-printing machine and PETG plastic. No money was spent on training: the team uses its own computers and the free Kaggle GPU quota, whose limits are covered in the risk plan.

2.5 Change Management Process

[Describe the process that will be followed for any changes in the project scope, schedule, resources, or other requirements.]

2.6 Closure and Evaluation

[Describe the process for closing and evaluating the project upon completion, including how experiences and lessons learned from the project will be summarized.]

n

III. Existing Systems

[Provide the final existing systems following the template as part II in the Report #3]

[This Section is usually titled "Literature Review" or "Related Work". For projects related to specific technologies or systems, you might call this section "Existing Systems" or "State of the Art". Choose a title that fits the content and context of your project.]

1. Overview of the Field

[Provide an overview of the field of study related to your project. This helps place your project within the broader context of that field.]

Robotics aims to relieve people of repetitive, tiring, or hazardous physical work by producing motion that is useful, reliable, and safe. Since the field took shape in the 1950s, it has drawn on mechanical and electrical engineering, rigid-body mechanics, control theory, and software. Any working system must connect planning software to the hardware that carries out motion. Fig. [X] illustrates the methods for motion generation, which are broadly categorized into two primary families:

+ *Explicit approaches* build on dynamics, using precise descriptions of a robot's rigid links and of their contact with the environment.
+ *Implicit approaches* treat motion as a statistical pattern to be learned from sensorimotor readings (Bekris et al., 2024).

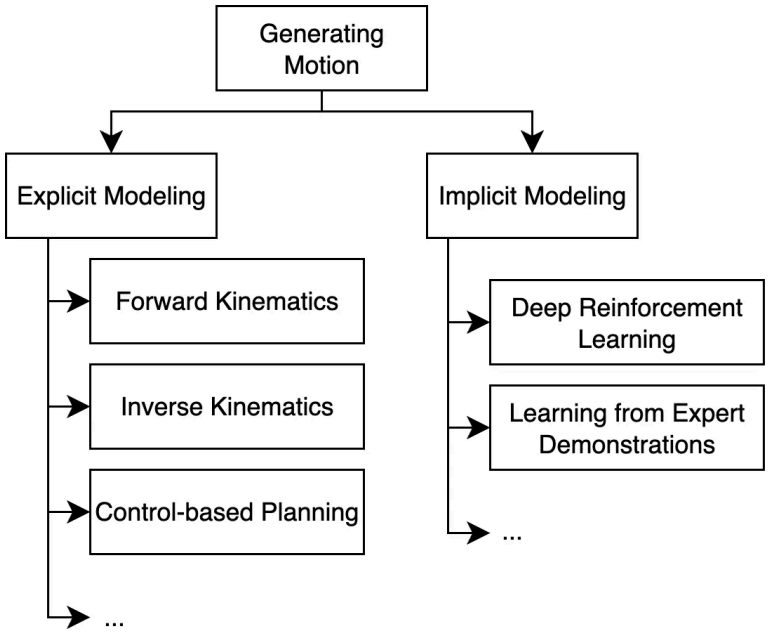


Figure … . Taxonomy of Motion Generation Methods

Robot learning belongs to the second family. Its rise reflects how hard it is to derive accurate models of robot–environment interaction, and recent results in locomotion, whole-body control, and manipulation suggest that learned behavior can rival explicit modeling (Kober et al., n.d.).

Learning-based methods offer several advantages over the classical stack. Visuomotor policies map sensor readings directly to actions, so the interfaces between separate perception, planning, and control modules disappear. Learned feature extraction readily absorbs different input modalities. Where dynamics are unknown or impractical to model, interaction data can replace an explicit model altogether. Robot learning can also draw on the growing supply of open robotics data, much as computer vision and natural language processing grew with large-scale corpora.

As illustrated in **Fig. [X]**, two main classes of methods have gained prominence: Behavioral Cloning (BC) and Reinforcement Learning (RL). On physical hardware, RL is hard to apply because exploration is costly and potentially unsafe, dense rewards are brittle and task-specific, and episodic training needs dependable resets. BC avoids these obstacles by casting control as imitation of expert demonstrations. The learned behavior then inherits the goals and success criteria implicit in the data, and no hand-crafted reward is needed. This project lies within BC and covers its two main branches. Single-task policies, represented by Action Chunking with Transformers (*ACT*), are typically not language-conditioned and each performs one task. Generalist policies, represented by *SmolVLA*, are conditioned on language instructions and produce motion across many tasks. The two branches differ in spirit but share a training principle, since both reproduce trajectories from a set of demonstrations, and this common principle justifies treating them together. The application studied here is tabletop manipulation, in which a robot arm changes the state of objects in its workspace through dexterous motion.

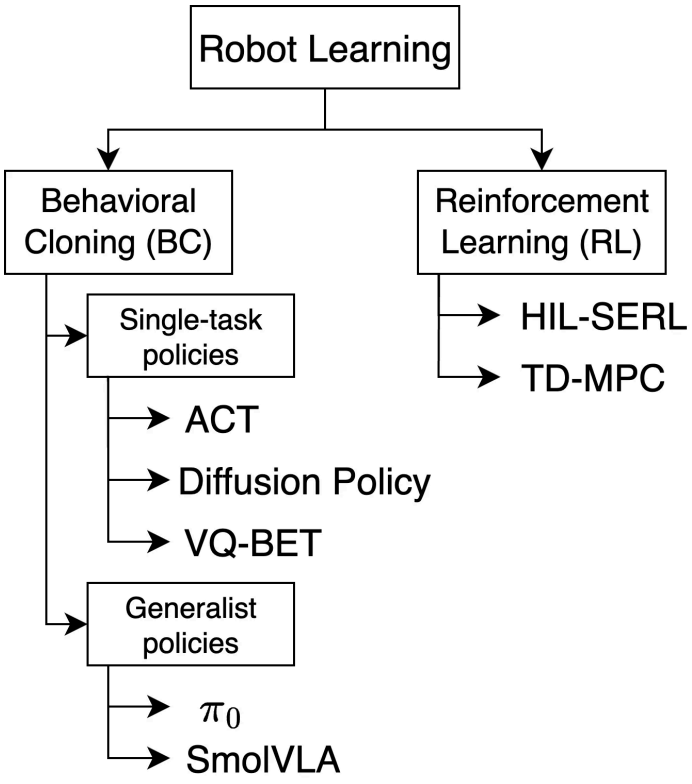


Figure … . Taxonomy of Robot Learning Methods

2. Historical Context

[Provide some historical background or developments in the field, especially those directly related to your project.]

BC has a long history, Pomerleau (1988) framed it as learning a mapping from observations to actions, a problem that can be approached through ordinary supervised learning. For decades afterward, robotics data stayed small and bound to individual tasks. Each expert trajectory was tied to one robot platform and the conditions of one environment, and human experts often performed the same task in quite different ways. Pooling such data was therefore difficult, and combining demonstrations from different embodiments could even cause negative transfer when control strategies conflicted. The result was a generation of specialist policies that performed well on their designated task but did not carry over to new deployments.

Two developments loosened these constraints. The first concerned hardware. ALOHA and ACT (Zhao et al., 2023) showed that fine-grained manipulation could be learned on low-cost, imprecise arms, which encouraged community-built teleoperation systems, open datasets, and the LeRobot library (see ***Fig. [X]*** for an overview of its end-to-end framework). The SO-ARM101 used in this project descends from that movement.

The second concerned scale. Large-scale models in natural language processing and computer vision had shifted those fields from task-specific training to a two-stage pattern of broad pretraining followed by task-specific adaptation, and robotics researchers began to ask whether the same recipe could work for control. Early generalist efforts such as BC-Z (Jang et al., 2022), RT-1 (Brohan et al., 2023), and RT-2 (Brohan et al., 2023a) were followed by open datasets, including Open X-Embodiment (O'Neill et al., 2025) and DROID (Khazatsky et al., 2025), that made open-source models possible. OpenVLA, π0, and SmolVLA are the main examples (Kim et al., 2024; Black et al., 2024; Shukor et al., 2025). In parallel, decentralized collection by individual researchers and hobbyists, supported by LeRobot, has become a significant source of data. Generalist models have also grown smaller and easier to run on limited hardware, a trend that matters for robots deployed outside large laboratories.

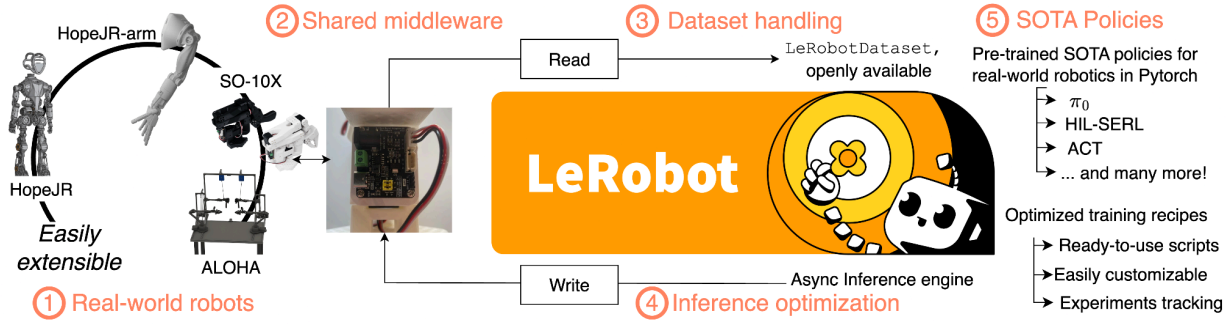


Fig … . LeRobot (An Open-Source Library for End-to-End Robot Learning)

3. Key Studies and Theories

[Describe and assess key studies, theories, and published works related to your topic. This includes both fundamental and applied research.]

The development of robot learning has progressed from task-specific imitation-learning policies toward generalist systems capable of understanding visual observations, interpreting natural-language instructions, and generating corresponding robot actions. Early approaches primarily focused on learning motor behaviours from demonstrations, whereas more recent Vision-Language-Action (VLA) models incorporate pretrained vision-language models (VLMs) to transfer semantic knowledge into robotic control.

3.1. Task-Specific Imitation Learning: Action Chunking with Transformers (ACT).

Action Chunking with Transformers (ACT) (Zhao et al., 2023) represents an important development in demonstration-based robot learning. Rather than predicting a single action at each time step, ACT employs a transformer-based policy to predict a sequence of future actions, referred to as an action chunk. Its architecture incorporates a Conditional Variational Autoencoder (CVAE) to model variations in demonstrated action sequences, while temporal ensembling combines overlapping predictions during inference. Action chunking reduces the effective prediction horizon and helps mitigate error accumulation during sequential execution.

Unlike VLA models, ACT does not incorporate natural-language instructions or a pretrained vision-language backbone. Instead, it learns a task-specific mapping from visual observations and robot states to motor commands using collected demonstrations. Its relatively straightforward architecture and demonstration-based training paradigm make it a useful reference for evaluating the benefits of language-conditioned robotic policies. In this project, ACT therefore serves as a conventional imitation-learning baseline against which the performance of SmolVLA-based approaches can be assessed.

3.2. The Emergence of Vision-Language-Action Models: BC-Z, RT-1, and RT-2.

The introduction of language-conditioned robotic policies marked a transition from learning individual manipulation behaviours toward developing more generalizable robotic systems. BC-Z demonstrated that robot policies could be conditioned on task descriptions, enabling a single policy to perform multiple manipulation tasks. Building on this direction, RT-1 (Brohan et al., 2023) expanded robotic learning through large-scale demonstrations collected across multiple robots and tasks. Its transformer-based architecture processes camera observations and natural-language instructions to predict discretized robot actions. This demonstrated the potential of scaling both demonstration datasets and policy architectures to improve task generalization.

RT-2 (Brohan et al., 2023a) further advanced this direction by introducing the Vision-Language-Action (VLA) paradigm. Rather than treating robotic control as an entirely separate prediction problem, RT-2 adapts pretrained vision-language models, including PaLI-X and PaLM-E, to generate robot actions within the same token-prediction framework used for language generation. By combining web-scale visual-language knowledge with robotic demonstrations, the model can transfer semantic understanding

to manipulation tasks involving objects and concepts that were not explicitly represented in its robotic training data. This approach illustrates the potential of pretrained multimodal representations to improve robotic generalization and establishes the VLM backbone as an important architectural component of VLA systems.

3.3. Large-Scale Robotic Datasets and Open VLA Architectures.

The development of generalist robotic policies has been closely associated with the availability of large and diverse demonstration datasets. Open X-Embodiment consolidated robotic datasets from multiple institutions and embodiments, providing approximately 1.4 million trajectories for studying cross-robot learning and transfer. Its findings demonstrated the potential advantages of training policies on diverse multi-embodiment data rather than relying exclusively on demonstrations collected from individual robotic platforms. Similarly, DROID introduced a large-scale dataset of real-world robotic interactions, further expanding the availability of demonstrations collected in diverse environments.

Despite these advances, access to large proprietary VLA systems and their training resources remains a barrier to independent research. OpenVLA addressed part of this limitation by releasing model weights, training procedures, and evaluation resources. It combines a pretrained vision encoder with a Llama 2 7B language model, projecting visual representations into the language model's embedding space before generating discretized robot actions. Although this architecture makes large-scale VLA research more accessible, its parameter count and computational requirements still present challenges for researchers working with limited hardware resources.

3.4. Lightweight Vision-Language-Action Models: SmolVLA.

SmolVLA (Shukor et al., 2025) addresses the computational accessibility challenges associated with large-scale VLA architectures. It introduces a compact and efficient framework designed to support robotic learning and deployment on more affordable hardware. The architecture combines a pretrained SmolVLM vision-language backbone with a dedicated action expert that generates continuous robot actions using flow matching. Several architectural optimizations, including visual token reduction, selective vision-model layer execution, and interleaved attention mechanisms, are employed to reduce computational overhead while retaining the benefits of pretrained multimodal representations.

Besides, SmolVLA introduces an asynchronous inference framework that separates action prediction from robot execution, improving responsiveness during real-world deployment. Its compatibility with affordable robotic platforms, including the SO-100 and SO-101, makes it particularly relevant to research conducted with low-cost manipulators. Nevertheless, reducing model size does not necessarily eliminate limitations associated with the representational capabilities of the underlying VLM backbone. The backbone determines how visual information and language instructions are encoded before being incorporated into the action-generation process. Consequently, its architecture and pretrained capabilities may influence the model's ability to distinguish objects, interpret spatial relationships, and understand task instructions.

3.5. SmolVLM2 - Lightweight Vision-Language Backbone.

SmolVLM2 (Marafioti et al., 2025) is a family of compact vision-language models developed to provide multimodal understanding capabilities with relatively low computational requirements. It supports visual and textual reasoning tasks, including image description, visual question answering, and document understanding. Its lightweight architecture makes it suitable for integration into resource-constrained applications, including robotic systems. However, most of it training data is for document reading and OCR tasks, which leads to it weakness in handle images. This raises the possibility of replacing SmolVLM2 in SmolVLA with alternative lightweight VLMs that may offer more suitable representations for robotic control while maintaining comparable computational efficiency.

This consideration motivates the present project to investigate alternative lightweight VLM backbones within the SmolVLA framework. Rather than developing an entirely new VLA architecture, the project examines whether replacing the original visual-language backbone with alternative compact multimodal models can improve task performance while maintaining computational efficiency on the SO-ARM101 platform.

4. Technological Advancements

[Discuss recent or significant technological advancements in your field, and how they affect your project.]

Three developments have made this project feasible on a student budget. Low-cost open hardware lowered the price of collecting robot data. An open software stack unified how that data is stored, used for training, and deployed. Compact vision-language models (VLMs) then reached a size where they can serve as backbones for lightweight policies. Each affects a different part of the project, from data collection to the design of the model.

Low-cost, open robot hardware. Early learning-based manipulation needed expensive, precisely engineered arms. ALOHA showed that fine-grained tasks could be learned on inexpensive teleoperated hardware, and ALOHA 2 later refined this approach (Aldaco et al., 2024). The SO-100 and SO-101 arms (Knight et al., 2024) extend it further. They use consumer-grade parts and 3D-printable shells, and the LeRobot authors put a single arm at roughly 225 euros, against about 21 thousand euros for the ALOHA bimanual platform. The arms come in leader–follower pairs. A person moves the leader arm, the follower copies the motion, and the recorded motion becomes training data. The same pair of arms therefore serves for both demonstration and deployment. This is the setup used in the project, and it is why a student team can collect its own task-specific demonstrations instead of relying only on public datasets.

1. An open software and data ecosystem: LeRobot. Hardware alone leaves a practical gap, because robot data has historically been scattered across incompatible formats and robot-specific control code. LeRobot (Cadene et al., 2026) addresses this with a single library covering the whole stack. A shared Python middleware reads the leader arm and drives the follower, whether under teleoperation or under a learned policy. The LeRobotDataset format stores sensor readings, multi-camera video, and task descriptions in one schema and can be streamed without downloading the full corpus. By September 2025, over 16 thousand datasets from more than 2.2 thousand contributors had been shared in this format, and over half of those datasets came from SO-10X arms. The library also provides tested PyTorch implementations of ACT, Diffusion Policy, π0, and SmolVLA, with training scripts for each. Its

inference stack separates action prediction from action execution, so a policy can run on a separate, more powerful machine and stream action chunks to the robot. In the authors' real-world tests on the SO-100, this asynchronous scheme gave similar success rates to synchronous inference but shorter cycle times, although the test used only ten episodes per task. For this project, LeRobot is the common platform on which all four model configurations will be trained, run, and compared under identical conditions.

2. Generative action models and compact VLAs. Modern policies no longer predict a single action per step. ACT predicts short chunks of future actions with a conditional variational autoencoder. Diffusion Policy and later flow-matching models such as π0 and SmolVLA learn the full distribution of actions, which suits demonstration data where experts act differently from one attempt to the next. Within this line, model size has become a design variable. LeRobot's measurements place ACT at about 52 million parameters and SmolVLA at about 450 million, compared with 3.5 billion for π0. In fp32, SmolVLA peaks at roughly 1.75 GB of GPU memory, while π0 needs about 13 GB. This makes a language-conditioned policy practical on a single consumer GPU and gives SmolVLA a clear role as the efficiency-oriented generalist baseline here.

3. Compact VLMs as backbones. The most recent change is in the VLMs that supply a VLA's visual and linguistic understanding. According to Li et al. (2026), VLM design has moved from separate vision and language towers, through projector-based adaptation of a pretrained language model, to natively multimodal trunks. Model scale has diverged along the way. Frontier systems have grown toward trillions of parameters, while a parallel line has shrunk to sub-billion models for edge deployment. Two members of this second line are central to this project. FastVLM (Vasu et al., 2025) pairs a hybrid convolution–transformer encoder, FastViTHD, with a small language model and controls the visual token count through input resolution alone, avoiding a separate pruning step. LFM2-VL, available at 450 million parameters, builds on a hybrid backbone of convolution and attention layers Both target low latency and a small memory footprint, the same constraints SmolVLA was designed around.

These developments set the project's boundaries. Open arms and LeRobot make it possible to collect data, train, and deploy within the team's own resources. Flow-matching VLAs with small action experts leave room to change the backbone without redesigning the policy. The newer compact VLMs offer candidate replacements that differ from SmolVLM in pretraining data and in how they handle visual tokens. Their published results come mostly from image-question answering, document, and chart benchmarks, and FastVLM's reported latencies were measured on Apple silicon. Whether these gains carry over to manipulation success and to inference speed on the project's hardware has not been established, which is the question the experiments are designed to answer. The LeRobot authors also acknowledge that quantization and graph compilation, which would further cut inference cost, are outside the library's current scope, so the efficiency figures measured here should be read as unoptimized baselines.

5. Comparison of Existing Systems

[If your project involves systems or technology, compare and contrast existing systems or technologies, including their advantages and disadvantages compared to your project.]

This section presents two comparisons to establish the motivation behind the proposed project. The first examines existing robot control policies, focusing on their architectures, capabilities, and computational requirements. The second compares robotic hardware platforms to justify the selection of SO-ARM101 as an affordable and accessible platform for VLA research. Together, these comparisons highlight the need for improving the capabilities of lightweight VLA models while maintaining compatibility with low-cost robotic hardware.

5.1. Comparison of Existing Robot Control Policies

Recent advances in robot learning have introduced different approaches to learning manipulation policies from demonstrations and multimodal observations. Among these approaches, Action Chunking with Transformers (ACT), π0, and SmolVLA represent different trade-offs between computational efficiency, task understanding, and general-purpose manipulation capabilities.

Aspect	**ACT**	**π0 (Pi-Zero)**	**SmolVLA**
Model category	Imitation learning policy	Generalist VLA	Lightweight VLA
Core architecture	Conditional VAE and Transformer	Pretrained VLM and flow-matching action expert	Pretrained VLM and flow-matching action expert
Language conditioning	Not inherently supported	Supported	Supported
Visual understanding	Learned visual features using CNN backbone	Pretrained vision-language representations	Pretrained vision-language representations
Action generation	Chunked action prediction	Continuous action generation through flow matching	Continuous action generation through flow matching
Model parameters	Approximately 80M	Approximately 3B	Approximately 450M

Main advantages	Computationally efficient, relatively easy to train, effective for fine-grained imitation learning	Strong general-purpose manipulation capabilities, multimodal understanding, and cross-task potential	Smaller architecture, open-source implementation, and reduced computational requirements
Main limitations	Limited language-based task understanding and dependence on task-specific demonstrations	High computational and data requirements compared with compact policies	Reduced model capacity and potential limitations inherited from its pretrained VLM backbone

Table … . Policy comparisons

ACT provides an efficient baseline for learning task-specific manipulation from demonstrations, but does not inherently offer the language-conditioned semantic understanding of VLA models. In contrast, π0 demonstrates the potential of large-scale vision-language-action architectures, although its computational requirements may limit accessibility for resource-constrained research environments. SmolVLA occupies an intermediate position by providing multimodal task understanding with a considerably smaller architecture. Nevertheless, the efficiency of SmolVLA does not necessarily guarantee optimal visual and semantic representations for every manipulation task. Since its action-generation component depends on representations produced by the pretrained VLM backbone, the choice of backbone may influence downstream policy performance.

Therefore, this project investigates whether replacing the original VLM backbone with an alternative lightweight vision-language model can improve the representations available for robotic control while maintaining reasonable computational requirements. This approach aims to investigate the potential to improve SmolVLA without introducing the computational overhead associated with substantially larger VLA architectures.

5.2. Comparison of Existing Robotic Hardware Platforms

The choice of robotic hardware is another important consideration in developing and evaluating VLA systems. Different robotic platforms provide different levels of mechanical capability, precision, accessibility, and financial requirements. This project considers Franka Emika Panda, ALOHA, and SO-ARM101 as representative platforms with different hardware designs and research objectives.

Aspect	Franka Emika Panda	ALOHA	SO-ARM101
Platform type	Single-arm research manipulator	Bimanual teleoperation system	Low-cost leader-follower robotic arm system
Configuration	Single arm	Two robotic arms	Leader and follower arms
Degrees of freedom	7	6 per arm in the original setup	5 + gripper per arm
Actuation	High-performance robotic actuators	Dynamixel-based actuators	Feetech bus servos
Main strengths	High precision, reliability, and advanced manipulation capabilities	Bimanual manipulation, teleoperation, and fine-grained demonstration collection	Low hardware cost, open-source design, and accessible teleoperation
Main limitations	Higher acquisition cost and hardware requirements	More complex setup and higher overall hardware cost than a single low-cost arm	Limited payload, mechanical rigidity, and precision compared with research-grade manipulators
Primary research suitability	Advanced manipulation and robotic control research	Bimanual imitation learning and complex manipulation	Accessible imitation learning and VLA experimentation
Image			

Table … . Hardware Comparison

Franka Emika Panda. The Franka Emika Panda is a seven-degree-of-freedom collaborative robotic arm designed for research and industrial applications. Its advanced actuation, reliable control interfaces, and high-precision manipulation capabilities make it suitable for investigating complex robotic tasks. However, its acquisition and maintenance costs are considerably higher than those of low-cost educational robotic platforms. This creates a financial and infrastructure barrier for smaller research teams and institutions seeking to conduct repeated real-world manipulation experiments.

ALOHA. ALOHA, introduced by Zhao et al. , is a low-cost, open-source bimanual teleoperation system designed for collecting demonstrations and learning fine-grained manipulation skills. It combines two robotic arms with a leader-follower teleoperation interface, allowing human operators to demonstrate coordinated manipulation tasks. The original system demonstrated the ability to perform challenging tasks, including cable-tie threading, object insertion, and other contact-rich manipulation activities. Although ALOHA significantly improves accessibility compared with conventional high-end robotic setups, its bimanual configuration requires more hardware components, space, and setup effort than a single-arm platform. Furthermore, its original hardware configuration was developed around higher-cost robotic components than those used in SO-ARM101.

SO-ARM101. SO-ARM101 is an open-source, low-cost robotic arm platform developed within the SO-ARM project and integrated with the Hugging Face LeRobot ecosystem [4]. Its leader-follower configuration allows users to collect demonstrations through direct teleoperation and subsequently use the collected data to train imitation-learning and VLA policies. The platform uses affordable bus servos and 3D-printed structural components, substantially reducing the hardware requirements compared with conventional research manipulators. Despite its limitations in payload capacity, mechanical rigidity, and positioning precision, SO-ARM101 provides sufficient capabilities for investigating fundamental tabletop manipulation tasks, including object picking, placing, and stacking. Its compatibility with existing machine-learning tools and open-source hardware design also makes it easier to reproduce experiments and modify the system for different research requirements.

These platforms demonstrate that meaningful robotic manipulation research does not necessarily require expensive, high-performance industrial hardware. Although Franka Panda provides advanced mechanical capabilities and ALOHA enables complex bimanual manipulation, their hardware configurations introduce additional financial and infrastructure requirements. SO-ARM101 offers a more accessible alternative for studying imitation learning and VLA-based control, particularly for tabletop manipulation tasks that do not require high payload capacity or advanced force control. Importantly, the purpose of selecting SO-ARM101 is not to demonstrate that it can replace more advanced research manipulators in every application. Instead, it provides a controlled and affordable environment for investigating whether lightweight VLA architectures can achieve reliable manipulation performance under practical hardware constraints.

By combining a low-cost robotic platform with an improved lightweight VLA policy, this project aims to contribute toward making VLA experimentation more accessible to researchers and educational institutions with limited resources.

6. Gaps in the Literature/Technology

[Identify gaps or limitations in the current research or technological systems, and how your project intends to address or improve upon these issues.]

In this section, we will talk about the research gaps we find in this SmolVLA paper, in which we will directly quote some sentences from their own paper as supporting evidence. In their limitations section, the author of SMOLVLA has mention about model size, the compact architecture make the model hard to scale and their cant be a trade off to improve, have to sacrifice the efficient for stronger performance. This is true, recent developments is leading toward to introduce new algorithms and models, but the more they add, the more weights their policy getting heavier. The table … below show a list of VLA policy and the compariosn between parameter of models published later than SmolVLA from June/2025 up to now.

Policy	**No.Parameters in Billions**	**Paper**	**Published date**
S^2-VLA	2B	*S^2-VLA: State-Space Guided Vision-Language-Action Models for Long-Horizon Manipulation*	2026-06-29 (arXiv:2606.27872)
FabriVLA	0.88B	*FabriVLA: A Lightweight Vision-Language-Action Model with Conformal Action Chunk Uncertainty*	2026-07-10 (arXiv:2607.08575)
PredVLA	~0.77M	*PredVLA: A Sub-Million-Parameter Predictive-Coding Policy for Robot Manipulation*	2026-08-27 (arXiv:2608.26673)
MINERVA	0.54M	*MINERVA: How Small Can a Manipulation Policy Be and Still Solve LIBERO?*	2026-09-03 (arXiv:2609.03715)
V-JEPA Policy	0.9B	*V-JEPA Policy: Building Effective World-Action Models on Predictive Visual Latents*	2026-09-29 (arXiv:2609.37250)
SPAR	1.12B	*SPAR: Structural Pruning and Action Recovery for Vision-Language-Action Models*	2026-10-19 (arXiv:2610.26487)
RynnVLA-Latent	4B	*RynnVLA-Latent: Scaling Robotic Manipulation Foundation Models with 3D-Grounded Latent Actions*	2026-10-25 (arXiv:2610.34356)

Table … . VLA model's parameter comparison

To the best of our knowledge there was no sign for VLA models smaller than SmolVLA, that is good enough to be published. This has proved their statement that the trade-off exists, making SmolVLA with 456 millions of parameters - the most lightweight policy at the model - a sota of its own efficiency and perfomance. This pose as gap for no solution solving this trade-off nor exist a smaller but better policy than SmolVLA.

From this scenario, we come up with an idea to leverage Smoval own design of architecture and improve it better instead of creating a complete new one, or as we say is adopt it architecture and make it better . One of the things we can change here is showin by it second limitations, choice of VLM backbones, state in its paper that the off-the-shell vlm model, SmolVLM, that it rely on is pretrain heavily for document reading and OCR tasks. The claims of missing task for handle images like identify objects, localization, … making it the best choice up-to-date as more and more new vlm are introduce.

We did not blindly believe in what the author of Smolvla so we take a look into smolvlm paper, "SmolVLM: Redefining small and efficient multimodal models". SmolVLM's vision pretraining pipeline is split into static image pretraining and video fine-tuning. **Static Vision Pretraining:** Nearly half of SmolVLM's image training mix (**48%**) is dedicated strictly to **OCR & Documents** (e.g., DocVQA, MathWriting, chart and table understanding). The remaining image allocation covers general captioning (14%), reasoning & logic (9%), table understanding (9%), chart understanding (12%), and visual Q&A (8%). **Video Fine-Tuning Data:** The video mixture focuses on high-level narrative and informational concepts—primarily **Information Seeking (60%)**, **Visual Description & Captioning (76% of video data)**, and **Temporal Understanding (18%)** sampled from passive datasets like LLaVA-Video, Video-STAR, Vista-400k, and FineVideo.

So, our solution for two solve these 2 problem is that we want to swap the vlm backbone to newer models, the run train and evaluate to see if the backbone swap does improve the perfomance while still mainstce the lightweight, efficient of smolvla.

Our research question is : **Can a carefully selected alternative VLM backbone improve manipulation performance while preserving, or further improving, SmolVLA's computational efficiency?**

The selected vlm choose as backbone replacement have to be lighiweight, it metrics are better than Smolvlm at some poinst. As these models are new and not natively inetrget into smolvla, we will develop an adapter algrohim to make them work together good.

7. Justification for the Project

[Based on the above information, highlight the necessity and uniqueness of your project. Explain why your approach to the problem is important and how it differs from or improves upon existing research or systems.]

[Note: This section is not just a list of studies but an analysis and evaluation part where you connect existing systems/research and works to your project, showing their linkage and impact on your work.]

IV. Methodology

[Provide the final methodology following the template as part II in the Report #4]

[This section will detail how you will approach the project, with a focus on applying and implementing AI techniques. The team is allowed to customize the content of this section, however, the team needs to provide a detailed view of the methods and techniques used. To ensure that the reader understands the scientific and systematic approach that you apply to your capstone project in the field of AI. Below are two sample templates for this section that you can choose from or customize.]

IV. Methodology (sample 1)

1. Research Questions and Objectives

[Identify the specific research questions and objectives that the project will address through the application of machine learning.]

2. Data Collection and Preprocessing

[Describe how you will collect data, including the sources of data, methods of collection, and any selection criteria. Information about the sample size, diversity, and representativeness of the data should be included.]

[Detail how the data will be processed before analysis, including cleaning data, handling missing data, transforming data, and normalization.]

3. Feature Selection and Engineering

[Describe the process of selecting and creating features that will be used for the machine learning model, as well as the reasons for these choices.]

4. Model Training and Validation

[Describe the machine learning models considered and the reasons for the final model choice, including technical specifications and configurations of the model.]

[Detail how the model will be trained and validated, including information on data splitting methods, cross-validation techniques, and the metrics used for evaluation.]

5. Evaluation Metrics

[Identify the metrics that you will use to assess the performance of the model, such as accuracy, F1 score, ROC AUC, etc.]

6. Implementation Plan

[Describe in detail how the model will be implemented, as well as any tools or technologies used to support this process.]

7. Ethical Considerations

[Discuss ethical issues that may arise when applying machine learning, including data privacy, accuracy, and how to handle data fairly.]

IV. Methodology (sample 2)

1. Project Objectives and RL Challenges

[Clearly identify the project objectives and the specific challenges that reinforcement learning aims to address, such as learning to optimize decisions in an uncertain environment.]

2. Environment Setup

[Describe the environment in which the reinforcement learning agent will operate, including parameters, states, and rewards.]

3. Agent Design and States

[Describe the design of the RL agent, including states, actions, and policies it may adopt.]

4. Reward System

[Detail the reward system, including how rewards and penalties are defined to guide the behavior of the agent.]

5. RL Algorithm

[Describe the reinforcement learning algorithm used, including any variations from standard methods such as Q-learning, Deep Q-Networks (DQN), Policy Gradients, etc. Mention how the algorithm will be applied to allow the agent to learn from experience.]

6. Training and Evaluation Strategy

[Describe the training and evaluation process for the agent, including how and how often the agent will explore and exploit the environment to learn. Information on how the performance and learning of the agent are tested in different situations should be included.]

7. Implementation Details

[Provide details on the implementation, including the software development environment, tools, and libraries used.]

8. Ethical and Safety Considerations

[Discuss ethical and safety issues related to the deployment and testing of the RL agent, especially in scenarios that might impact the real world.]

[Note: By adjusting and supplementing the content appropriately for the "Methodology" section, you will help ensure that this part accurately reflects the approach and specific challenges of the AI capstone project.]

V. System Design and Implementation

[Provide the final system design and implementation following the template as part II in the Report #5]

1. AI Model Integration

[Describe how the AI model is integrated into the larger system, including the interface and data exchange between the AI model and other system components.]

2. Data Flow and Processing

[Illustrate and describe the data flow through the system, from data collection, processing, to data analysis and result presentation.]

3. Deployment Strategy

[Detail how the system will be deployed, including the deployment environment and deployment procedures.]

4. Scalability and Maintenance

[Discuss how the system is designed to handle growth and future maintenance, including system updates, scalability, and version management.]

[Note: By providing a detailed view of the AI solution's design and implementation in this section, you will help readers understand the technical and systematic approach you have applied to your capstone project.]

VI. Results and Discussion

[Provide the final results and discussion following the template as part II in the Report #6]

1. Results and Analysis

[Present the results obtained from executing the project, including numerical data, graphs, diagrams, and supportive imagery. This may include the outcomes of machine learning models, such as accuracy, confusion matrices, ROC curves, and other evaluation metrics.]

[Provide a detailed analysis of the results obtained, explaining their significance and impact on the project. This might include comparing the performance of different models, error analysis, and interpreting the features of the model.]

2. Discussion

[Discuss the results, including how they address the research questions and hypotheses, their alignment with results from related studies, and the broader context of the project. Mention any limitations or challenges encountered during the project.]

3. Recommendations

[Offer recommendations based on the findings, including suggestions for further research, system improvements, or practical applications.]

4. Personal Reflection (Optional)

[A section reflecting personally on the project process, lessons learned, and how improvements could be made in the future.]

[Note: The "Results and Discussion" section is an essential part of the capstone project document, where you evaluate the success of your project based on data and analysis. This is also an opportunity for you to reflect and discuss the larger significance of the work you have done, as well as to identify any opportunities for further research or development.]

VII. Conclusion

[Provide the final conclusion following the template as part II in the Report #7]

1. Summary of Findings

[Summarize the main findings of the project, including how the research questions have been answered and how the project goals were achieved.]

2. Contributions and Reflections on the Project

[Highlight the specific contributions of the project to the field of study, community, or industry. This could include technical advancements, new knowledge, or practical applications.]

[Reflect on the project process, including what worked well and what could be improved. This is also an opportunity to acknowledge the level of challenge and learning from the project.]

[Provide some final thoughts or conclusion, expressing hopes or vision for the future of the research or the problem addressed by the project.]

3. Limitations and Future Work

[Acknowledge any limitations of the project and how they may affect the results and conclusions. This includes factors such as data, methods, or assumptions.]

[Suggest directions for future research or development based on the results and limitations of the project. This helps guide others who may build upon your work.]

[Note: The "Conclusion and Future Work" section is not only a place to summarize the completed work but also an opportunity to highlight the importance of the project and propose next steps. This helps readers understand the impact and potential of the work you have done.]

VIII. References

[Provide the final references following the template as part II in the Report #8]

[Formatting: Ensure that all references are listed in a uniform and correct format, following the guidelines of the citation system you have chosen (e.g., APA, MLA, Chicago, etc.). This helps ensure accuracy and ease of source verification.]

[Order: Typically, reference items are arranged in alphabetical order for easy lookup. Make sure that all sources are listed comprehensively and without omission.]

[Completeness: For each reference, provide enough information so that readers can find and review the source material. This includes the author's name, year of publication, title of the work, publisher name or journal name, and other relevant information such as page numbers or URL.]

[Accuracy: Carefully check to ensure that all information about the references is correct and up-to-date.]

[Relevance: Only include sources that were actually cited or used in your project. This helps maintain the integrity and accuracy of the project.]

[Note: The "References" section is an indispensable part of any academic project. It not only proves that you have conducted careful and grounded research but also allows future generations of students to check and expand their projects from the sources you have cited.]

IX. Appendices

[Provide the final appendices following the template as part II in the Report #9]

[If there is any supplementary information that does not fit into the main sections of the document but is still important and useful for a full understanding of the project, you can include it in the appendices. This might include raw data, source code, full versions of surveys, more detailed diagrams, or other supporting documents for your main content.]

[Note: After including all these sections, ensure that all parts of the document are logically and coherently organized, and that you have thoroughly checked for language, spelling, and grammar errors.]